

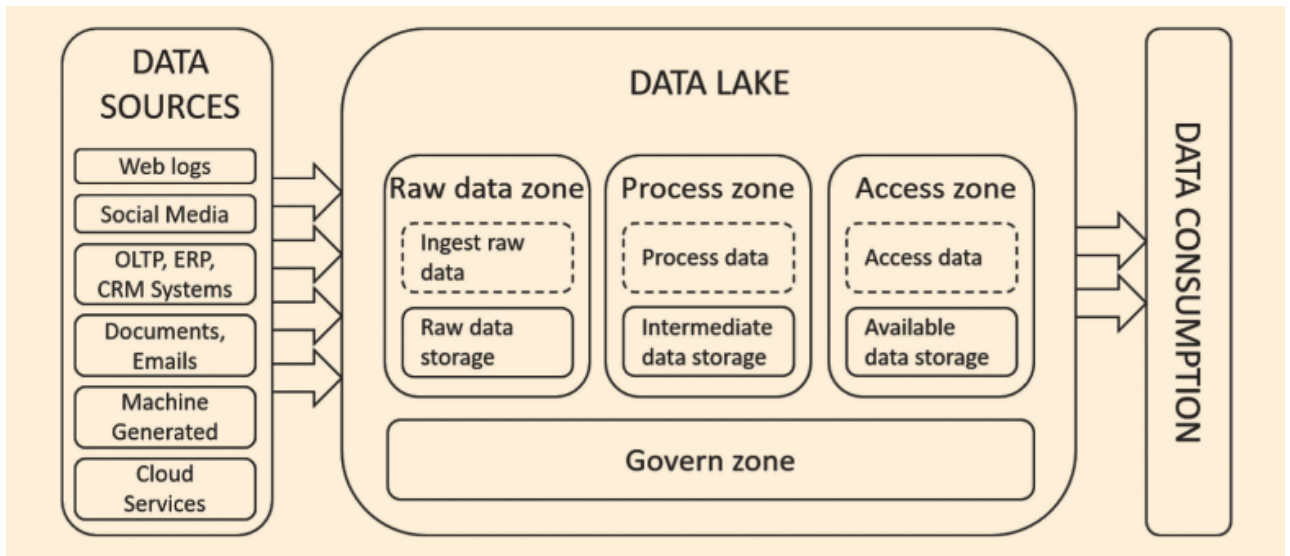


Figure 1: Architecture of a data lake

- **File-based storage systems:** Hadoop Distributed File System (HDFS) and Azure Data Lake by Microsoft are file-based data lakes used for storage.
 - **Single data store:** These types of data stores focus on specific types of data and use a single database system for their storage.
 - **Cloud-based data lakes:** Large scale commercial data lakes are available on cloud infrastructure like Amazon Web Services (AWS), Azure Data Lake Store, Google Cloud Platform (GCP), Alibaba Cloud, and the Data Cloud from Snowflake.
 - **Flexibility and agility:** Data lakes have a schema-on-read approach; so they can read any data type and format. Thus, data lakes enable a wider range of analyses than traditional decision-oriented databases, such as data warehouses and data marts, and show better flexibility and agility.
 - **Real-time data ingestion:** Data is ingested into the data lake without any transformation, minimising the time lag between data extracted from sources and its ingestion into databases.
 - **High scalability:** As data lakes are implemented using distributed technologies, they provide high scalability.
 - **Fault tolerance:** The underlying technologies of data lakes provide high resilience to both hardware and software failures, resulting in excellent fault tolerance.
- To sum up, data lakes empower organisations to uncover valuable insights from their data, paving the way for data-driven decision-making in the digital age. **END** 🐧

Advantages of data lakes

Data lakes empower organisations to gain insights and create actionable strategies. However, there is a lot more to them.

- **Cost-effective:** Data lakes are less expensive to deploy than traditional decision-oriented databases.
- **Data fidelity:** They preserve the original data to avoid any data loss that could occur from data preprocessing and transformation operations. However, data fidelity also introduces a high risk of data inconsistency in data lakes due to data integration from multiple, disparate sources without any transformation.

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